

INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY

HYDERABAD

TOPIC:-Automated Weapon Detection in Banks and ATMs Using YOLOv8

Introduction

As crime rates and armed robberies continue to rise, there is an urgent need to enhance security in critical locations like banks and ATMs. Traditional surveillance systems, which rely heavily on human operators, can be prone to errors and delays due to factors like fatigue and limited attention span. This project addresses these challenges by developing an automated weapon detection system using the YOLOv8 model. Designed to detect potential threats, such as knives, handguns, and rifles, in real-time video feeds, this system offers faster, more reliable threat identification, improving security and enabling a quicker response to dangerous situations.

Objective

To develop an automated, real-time weapon detection system using YOLOv8 to enhance security in banks and ATMs by identifying potential threats such as firearms and knives.

Project overview

This system utilizes a custom-trained YOLOv8 model deployed in Visual Studio Code to detect weapons in video feeds, providing immediate threat identification with high accuracy.

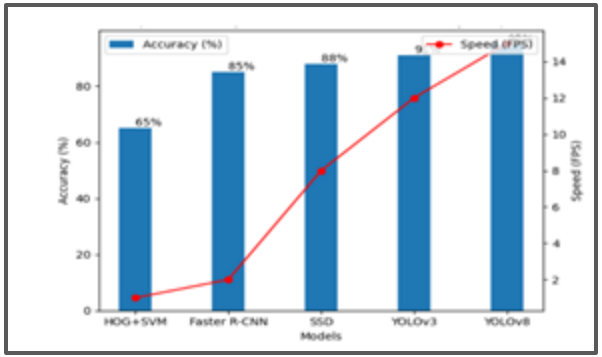
Challenges

Adapting the model to handle variations in lighting, camera angles, and complex scenes to reduce false positives and improve detection reliability.

Literature review

Study	Method	Strengths	Limitations
Dalal & Triggs (2005)	HOG + SVM	Effective for specific objects, e.g., pedestrians	High computational cost; not real-time
Ren et al. (2015)	Faster R-CNN	High accuracy, integrated region proposal	Slower, not ideal for real-time
Liu et al. (2016)	SSD	Faster, suitable for real-time applications	Limited precision with small objects
Redmon et al. (2016)	YOLO (Original)	Real-time performance, single regression task	Lower accuracy in dense scenes
Recent Studies (2021-2023)	YOLOv4, YOLOv5, YOLOv8	Enhanced speed and accuracy, adaptable architecture	Complex tuning for specific settings

Model	Accuracy(%)
Histogram of Oriented Gradients(HOG)+SVM	60-70
Faster R-CNN	70-85
Single Shot Detector(SSD)	75-85
YOLO	75-90
YOLOv5	85-92
YOLOv8	88-95



YOLOv8

- YOLOv8 is the latest model in the YOLO series, optimized for faster and more accurate object detection.
- Enhanced architecture increases both speed and accuracy, making it ideal for real-time security applications.
- Lightweight design allows it to run efficiently across devices, from powerful servers to mobile devices.

Advantages in Weapon Detection:

Real-time processing

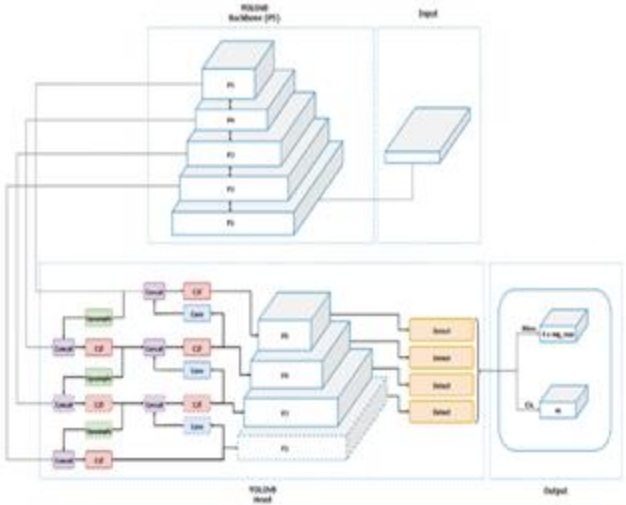
Can quickly analyze video frames, detecting threats instantly in high-risk environments like banks and ATMs.

High precision

Reduces false positives by accurately distinguishing weapons from other objects.

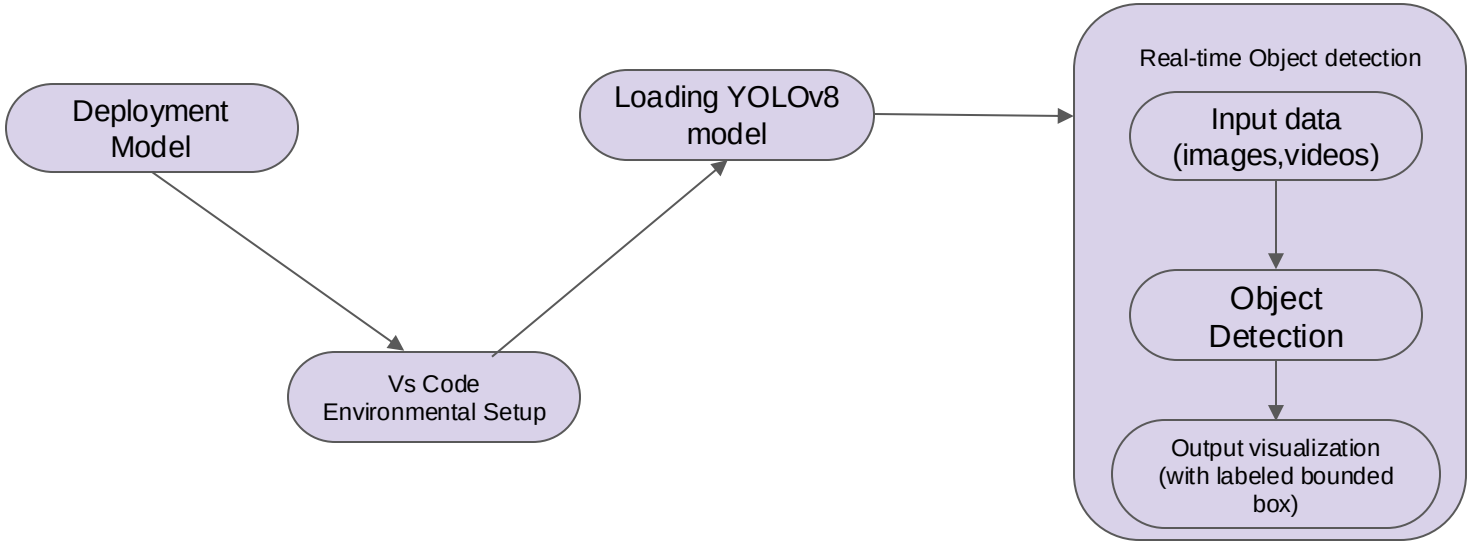
Versatile deployment

Efficient enough to work on various hardware setups, expanding its use in different security applications.



Methodology

1. **Data Collection:** Compiled custom dataset with weapon images (knives, pistols, rifles).
2. **Data Annotation:** Used Label-Studio to label images with bounding boxes and weapon types.
3. **Model Training:** YOLOv8 trained on Google Colab using customized parameters.
4. **Deployment:** Real-time detection implemented in Visual Studio Code.



Weapon detection dataset overview

Data Collection

- **Dataset**: Images of *Pistols*, *Machine Guns*, *Knives*, and non-weapon objects.
- **Sources**: Public domains, YouTube, CCTV footage, GitHub, University of Granada, and internet movies.
- **Data Split**: 80% for training, 20% for validation.
- **Challenges**: Required extensive, time-consuming data gathering as no dedicated dataset existed.

Data Classes

- **Pistol**: Images with handguns for identifying personal/security firearms.
- **Machine Guns**: Automatic firearms with high capacity, mostly military.
- **Knives**: Bladed objects like combat knives, close-range or hidden.
- **Non-Weaponed Elements**: Includes common objects to minimize false positives.

Training Benefits

- Covers varied threat types, from small concealed objects to large firearms.
- Enhances detection system's efficiency through class segmentation.

Data Annotation

We used Label-studio, an open-source tool, to annotate the dataset by drawing bounding boxes around weapons, labeling each as *knife*, *handgun*, or *long gun*. Annotations were saved in YOLO format (TXT) for compatibility with the YOLOv8 model, ensuring accurate bounding box placement for effective model training.

Image Class	Input Image	Annotated Image (Label-Studio)
Machine Gun		
Non-Weaponed Element		
Pistol		
Knives		

YOLOv8 Model Training and Deployment

Training on Google Colab

- **Setup**: Configured Colab with GPU resources and installed `ultralytics` for YOLOv8.
- **Data Preparation**: Uploaded annotated images and labels; configured model as `yolov8n.yaml` with dataset-specific adjustments.
- **Training Process**: Trained over 30 epochs with data augmentation (flipping, resizing) to improve performance. Saved best model as `best.pt`.

Deployment in Visual Studio Code

- **VS Code Environment**: Created Python virtual environment; installed dependencies for YOLOv8.
- **Loading & Detection**: Loaded `best.pt` model in a Python script for real-time detection on images/video streams.
- **Real-Time Object Detection**: Script displayed bounding boxes around detected weapons, enabling fast, accurate analysis in visual inputs.

Result

The YOLOv8 model excelled in detecting weapons in both images and video, achieving over 80% mAP on the validation set. It accurately identified knives, handguns, and long guns with minimal latency, making it suitable for live surveillance.

Detection Example →

		
Case 1: Pistol detected with 68.6% confidence.	**Case 2:** Knife detected with 76.6% confidence.	**Case 3:** Gun detected with 55.0% confidence.

These results demonstrate the model's ability to accurately detect weapons in various scenarios, highlighting its robustness for real-world applications.

Conclusion

The automated weapon detection system using the YOLOv8 model demonstrate for enhancing security in sensitive environments like banks and ATMs. Its high accuracy and real-time detection capabilities enable swift threat identification and response, reducing the need for continuous human monitoring.

Key Features of YOLOv8 Automated Weapon Detection System

- **Custom Datasets Development:** Tailored datasets boost model effectiveness.
- **Accessible Training Platforms:** Uses Google Colab for efficient training.
- **Future integration:** Building a unified security ecosystem.
- **Expanded Capabilities:** Detecting a broader range of threats.
- **User-Friendly Interfaces:** Ensuring practical utility for security personnel.

References

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